

SINA TABAKHI

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CURRENT POSITION

- **Postdoctoral Researcher in AI for Health**, The Institute of Cancer Research, London, UK

RESEARCH INTERESTS

- **Techniques:** Multimodal AI, graph neural networks, foundation models, feature selection
- **Applications:** AI for health, synthetic lethal vulnerability discovery, digital pathology, multiomics data analysis

EDUCATION

- **PhD in Computer Science**, University of Sheffield, UK Nov 2021 - Jan 2026
Thesis: Multimodal learning with graphs for multiomics cancer classification
Supervisor: Professor Haiping Lu
- **MSc in Computer Engineering**, University of Kurdistan, Iran Sep 2011 - Oct 2013
GPA: 19.45/20 (4.0/4.0); *Rank:* 1/20
- **BSc in IT Engineering**, Azad University, Iran Sep 2007 - Sep 2011
GPA: 18.41/20 (3.92/4.0); *Rank:* 1/80

PUBLICATIONS

Journal Papers

1. P. Tripathi, **S. Tabakhi**, M. N. Suvon, L. Schöb, S. Alabed, A. Swift, S. Zhou, and H. Lu, “Interpretable multimodal learning for cardiovascular hemodynamics assessment”, *IEEE Transactions on Medical Imaging* (**IF: 9.8**), 2026
2. X. Liu, J. Zhang, S. Zhou, T. L. van der Plas, A. Vijayaraghavan, A. Grishina, M. Zhuang, D. Schofield, C. Tomlinson, Y. Wang, R. Li, L. van Zeeland, **S. Tabakhi**, et al., “Towards deployment-centric multimodal AI beyond vision and language”, *Nature Machine Intelligence* (**IF: 23.9**), 7, 1612-1624, 2025
3. X. Liu, S. Rastegari, Y. Huang, S. C. Cheong, W. Liu, W. Zhao, Q. Tian, H. Wang, Y. Guo, S. Zhou, **S. Tabakhi**, et al., “Interpretable multimodal learning for tumor protein-metal binding: Progress, challenges, and perspectives”, *Methods* (**IF: 4.3**), 242, 97–112, 2025
4. **S. Tabakhi**, M. N. Suvon, P. Ahadian, and H. Lu, “Multimodal learning for multi-omics: A survey”, *World Scientific Annual Review of Artificial Intelligence*, 1, 2250004, 2023
5. **S. Tabakhi** and P. Moradi, “Universal feature selection tool (UniFeat): An open-source tool for dimensionality reduction”, *Neurocomputing* (**IF: 6.5**), 535, 156-165, 2023
6. **S. Tabakhi** and P. Moradi, “Relevance–redundancy feature selection based on ant colony optimization”, *Pattern Recognition* (**IF: 7.6**), 48, 2798-2811, 2015, (**226 citations**, May 2026)
7. **S. Tabakhi**, A. Najafi, R. Ranjbar, and P. Moradi, “Gene selection for microarray data classification using a novel ant colony optimization”, *Neurocomputing* (**IF: 6.5**), 168, 1024-1036, 2015, (**192 citations**, May 2026)
8. **S. Tabakhi**, P. Moradi, and F. Akhlaghian, “An unsupervised feature selection algorithm based on ant colony optimization”, *Engineering Applications of Artificial Intelligence* (**IF: 8.0**), 32, 112-123, 2014, (**520 citations**, May 2026)

Conference Papers

1. **S. Tabakhi**, C. Vandermeulen, I. Sudbery, and H. Lu, “Heterogeneous graph attention network improves cancer multiomics integration”, 33rd *International Conference on Intelligent Systems for Molecular Biology (ISMB/ECCB)*, Liverpool, UK, 2025 (Abstract accepted)
2. J. Zhang, X. Liu, W. Wu, **S. Tabakhi**, W. Fan, S. Zhou, K. L. Tee, T. S. Wong, and H. Lu, “Classifying the stoichiometry of virus-like particles with interpretable machine learning”, 47th *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Copenhagen, Denmark, 1-7, 2025
3. **S. Tabakhi** and H. Lu, “Multi-agent feature selection for integrative multi-omics analysis”, 44th *International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, Glasgow, UK, 1638-1642, 2022

Preprints

1. **S. Tabakhi**, C. Chen, and H. Lu, “Missing-modality-aware graph neural network for cancer classification”, *arXiv preprint* [arXiv.2506.22901](https://arxiv.org/abs/2506.22901), 2026
2. S. Rastegari, **S. Tabakhi**, X. Liu, T. Jiang, W. Sang, and H. Lu, “Co-evolution-based metal-binding residue prediction with graph neural networks”, *arXiv preprint* [arXiv.2502.16189](https://arxiv.org/abs/2502.16189), 2026
3. **S. Tabakhi**, C. Vandermeulen, I. Sudbery, and H. Lu, “Heterogeneous graph attention network with joint feature selection for cancer multiomics integration”, *arXiv preprint* [arXiv.2408.02845](https://arxiv.org/abs/2408.02845), 2024

HONORS AND AWARDS

- **Gold Reviewer Award**, International Conference on Machine Learning (ICML), 2026
- **Best Pitch Award**, First Workshop on Multimodal AI, Sheffield, UK, 2023
- **Runner Up Prize**, Open Research Prize 2023, University of Sheffield, UK, 2023
- **Best PGR ORCID Profile Prize**, PGR ORCID Profile Competition 2023, University of Sheffield, UK, 2023
- **PhD University Research Scholarship**, University of Sheffield, UK, 2021-2025
- **Top 5 Most Cited Papers** in Last 5 Years, Engineering Applications of Artificial Intelligence (Elsevier), 2019
- **Outstanding Contribution in Reviewing**, Pattern Recognition (Elsevier), 2018
- **Ranked 1st** among 20 students, Department of Computer Engineering, University of Kurdistan, Iran, 2013
- **Ranked 1st** among 80 students, Department of Computer Engineering and IT, Azad University, Iran, 2011
- **Honorable Mention** in the Asia Regional ACM Programming Contest, Tehran Site, Sharif University of Technology, Iran, 2008, 2009, and 2010
- **Ranked 1st** in Internal ACM Programming Contest, Azad University, Iran, 2008 and 2009

WORK EXPERIENCE

- **Software Engineer** (part-time), University of Sheffield, Sheffield, UK Jan 2022 – Oct 2023
 - Developed the Sheffield Data Science and AI Community website (<https://shef-ai.github.io/>)
- **Senior Backend Developer**, Phoenix, Tehran, Iran Dec 2019 – Oct 2021
 - Developed a crowdfunding platform using .Net Core, CQRS, Entity Framework, event sourcing, and SQL Server
 - Designed unit tests to cover 30% of the code
 - Refactored and maintained legacy code for the back-office project
 - Collaborated with front-end developers, infrastructure teams, and support staff to enhance project execution
- **Senior Backend Developer**, AloPeyk, Tehran, Iran May 2019 – Nov 2019
 - Developed micro-services based on .Net Core, Entity Framework, and MySQL for the AloMarket project
 - Implemented unit and integration tests, achieving 50% code coverage
 - Contributed to development meetings and architectural design discussions with the development team
- **Web Application Developer**, Jiro Software Engineering Co., Sanandaj, Iran Jan 2017 – May 2019
 - Contributed to academic software development, addressing bugs, and implementing data visualization features
 - Worked with ASP.Net, C#, and SQL server
 - Employed DevExpress and Stimulsoft tools for data visualization
- **Research Assistant**, University of Kurdistan, Sanandaj, Iran Jun 2015 – Apr 2016
 - Explored numerous state-of-the-art feature selection methods and implemented them in a Java-based package

TEACHING EXPERIENCE

- **Co-supervisor of an Intern**, University of Sheffield, UK 2024–2025
 - Project: Graph neural networks for protein-metal binding prediction
 - Results: Two journal papers (one accepted in *Methods*, one under review) and one conference paper (abstract accepted at ISMB/ECCB 2025)
- **Graduate Teaching Assistant**, Scalable Machine Learning, University of Sheffield, UK 2023-2024
 - Used PySpark for large-scale machine learning on the university’s HPC cluster systems
- **Instructor**, Design and Analysis of Algorithms, University of Kurdistan, Iran 2015
- **Instructor**, Data Structures, University of Kurdistan, Iran 2015

SELECTED RESEARCH PROJECTS

- **Universal Feature Selection Tool (UniFeat)** (<https://unifeat.github.io/>)
 - Developed an open-source tool in Java, incorporating 30 feature selection methods
 - Implemented auxiliary tools for performance evaluation, visual displays, and statistical analysis
 - Created comprehensive end-user documentation, including installation instructions and use case examples
 - Provided detailed API documentation and tutorials for developers to facilitate software extension
- **Knowledge-Aware machine LEarning library (PyKale)** (<https://pykale.github.io/>)
 - Contributed to the library, focusing on multimodal learning to enhance machine learning accessibility
 - Developed a multiomics integration method for cancer classification tasks
 - Designed unit and integration tests that increased the code coverage of the library
 - Contributed to developing API documentation and instructional examples

PROFESSIONAL SERVICES

- **Organizer** of the EMBC Workshop on [Open Biomedical Multimodal AI Research: From Pixels to Molecules](#), Copenhagen, Denmark, 2025
- **Organizer** of the [Third Workshop on Multimodal AI](#), London, UK, 2025
- **Organizer** of the [Second Workshop on Multimodal AI](#), Sheffield, UK, 2024
- **Organizer** of the [First Workshop on Multimodal AI](#), Sheffield, UK, 2023
- **Co-organizer** of the Alan Turing Institute’s interest group on [Meta-Learning for Multimodal Data](#), 2022-2025
- **Reviewer for journals:** Nucleic Acids Research, IEEE Transactions on Neural Networks and Learning Systems, IEEE Journal of Biomedical and Health Informatics, Pattern Recognition, Engineering Applications of Artificial Intelligence, Neural Computing and Applications
- **Reviewer for conferences:** International Conference on Machine Learning (ICML 2026), Learning on Graphs (LoG 2024), IEEE International Symposium on Biomedical Imaging (ISBI 2023)

TALKS AND POSTERS

- “Missing-modality-aware graph neural network for cancer classification”, *Third Workshop on Multimodal AI*, London, UK, Sep 2025
- “Heterogeneous graph attention network improves cancer multiomics integration”, *33rd International Conference on Intelligent Systems for Molecular Biology*, Liverpool, UK, Jul 2025
- “From multimodal learning to graph neural networks”, *First Workshop on Multimodal AI*, Sheffield, UK, Jun 2023
- “Multi-agent feature selection for integrative multi-omics analysis”, *44th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, Glasgow, UK, Jul 2022
- “Multi-agent feature selection for integrative multi-omics analysis”, *N8 CIR Machine Learning Theme Launch*, University of Leeds, UK, Nov 2022

MEMBERSHIPS

• The Alan Turing Institute, London, UK	2022-2025
• Insigneo Institute (Interdisciplinary research center revolutionizing healthcare), Sheffield, UK	2023-2025
• IEEE Member	2012-2025
• IEEE Engineering in Medicine and Biology Society Member	2022-2025

TECHNICAL SKILLS

- **Programming Languages:** - Experienced in Python, C# - Worked with Java, C/C++
- **Frameworks:** - Worked with PyG, PyTorch, PySpark, NumPy, Pandas, Scikit-learn, Matplotlib
- **Web Programming:** - Experienced in ASP.NET - Familiar with HTML, CSS, Javascript
- **Databases:** - Experienced in SQL server - Worked with MySQL
- **Version Controls:** - Git